



Computer
Science

CSC380: Principles of Data Science

Probability 3

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Review: “probability cheatsheet”

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Additivity:

For any *finite* or *countably infinite* sequence of disjoint events $E_1, E_2, E_3, \dots$,
$$P\left(\bigcup_{i \geq 1} E_i\right) = \sum_{i \geq 1} P(E_i)$$

Inclusion-exclusion rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Law of total probability: For events $B_1, B_2, \dots$ that partitions Ω ,

$$P(A) = \sum_i P(A \cap B_i)$$

Conditional probability:

$$P(A|B) := \frac{P(A \cap B)}{P(B)}$$

$(P(A|B) \neq P(B|A) \text{ in general})$

Probability chain rule:

$$P(A \cap B \cap C) = P(A|B \cap C)P(B|C)P(C)$$

Law of total probability + Conditional probability:

$$P(A) = \sum_i P(A \cap B_i) = \sum_i P(B_i)P(A|B_i) = \sum_i P(A)P(B_i|A)$$

Bayes' rule:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

Independence:

(definition) A and B are independent if $P(A, B) = P(A)P(B)$

(property) A and B are independent if and only if $P(A|B) = P(A)$ (or $P(B|A) = P(B)$)

- Random variables
- Distribution functions
 - probability mass functions (PMF)
 - cumulative distribution function (CDF)

Random Variables

Random variables (RVs)

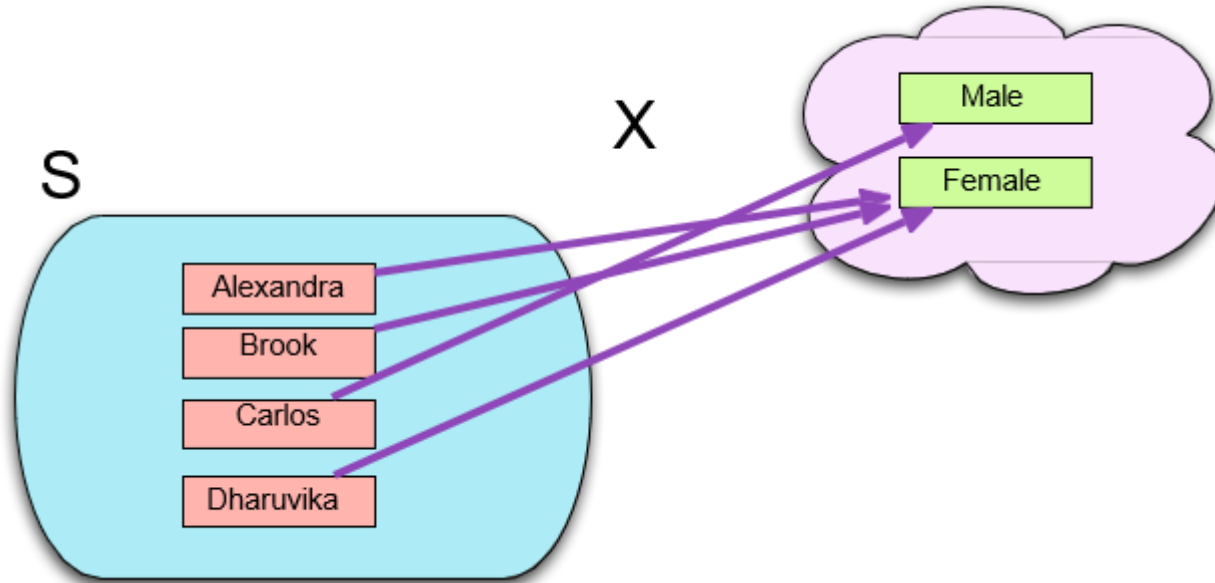
- A single random sample may have more than one characteristic that we can observe (i.e., it may be bi-/multivariate data).
- We can represent each characteristic (e.g., gender, weight, cancer status, etc.) using a separate random variable.

Random Variable

A **random variable** connects each possible outcome in the sample space to some property of interest.

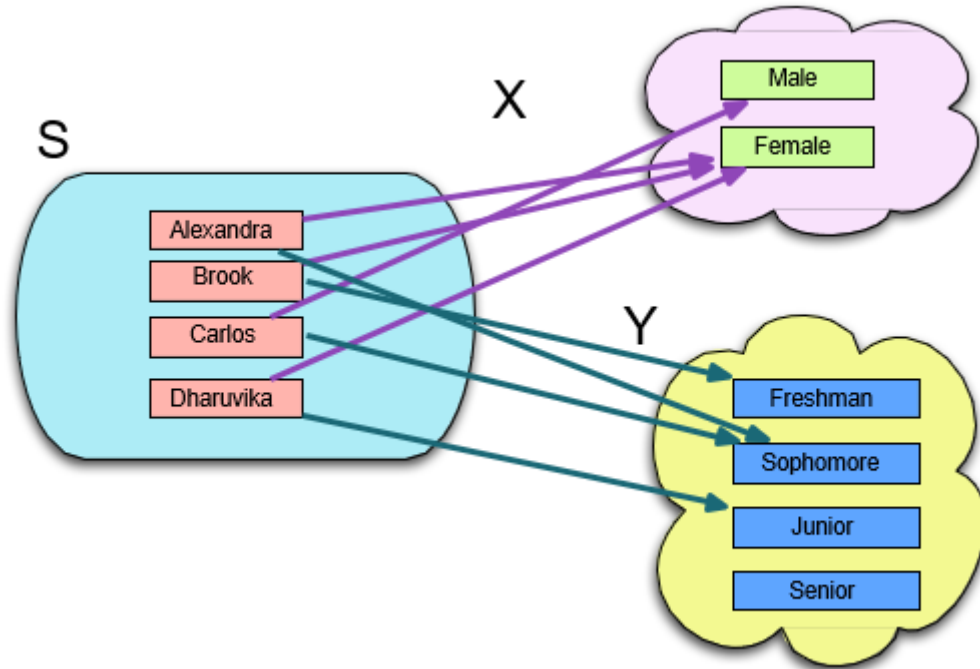
Each value of the random variable (e.g., male or female) has an associated probability.

Random Variable: Example



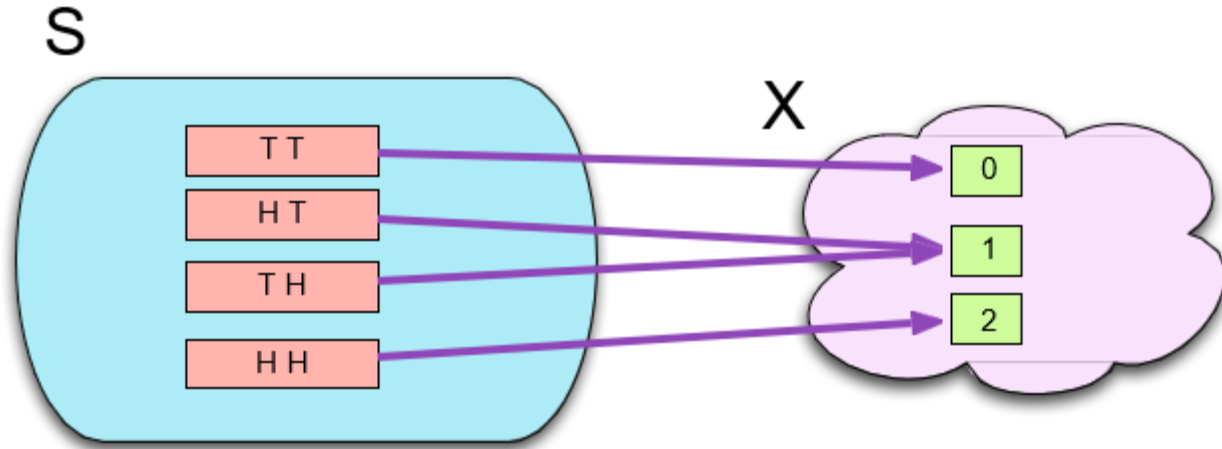
- X : people $\rightarrow$ their genders

Random Variable: Example



- Y : people $\rightarrow$ their class year

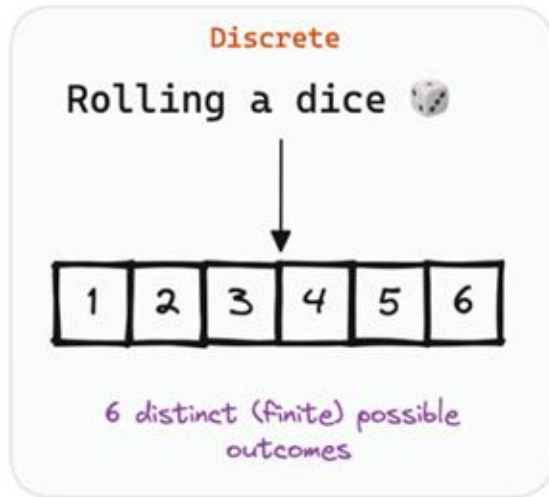
Random Variable: Example



- X : sequence of coin flips $\rightarrow$ Number of heads

Types of Random Variables

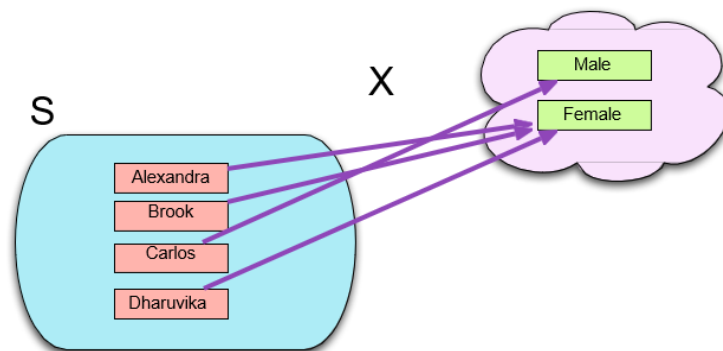
- Discrete random variable: takes a finite or countable number of distinct values.
- Continuous random variable: takes an infinite number of values within a specified range or interval.



Distribution functions

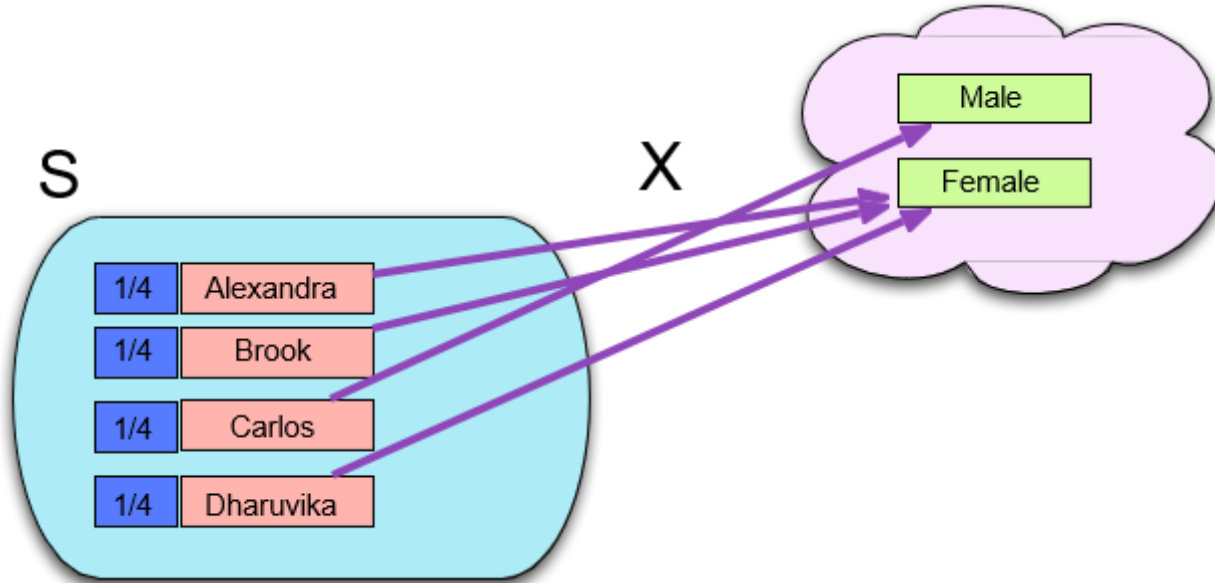
Discrete distributions

- When a random variable is discrete, its *distribution* is characterized by the probabilities assigned to each distinct value.
- The probability that the random variable takes a particular value comes from the probability associated with the set of individual outcomes that have that value.
 - This set is an event
- E.g. $P(X = \text{Female})$



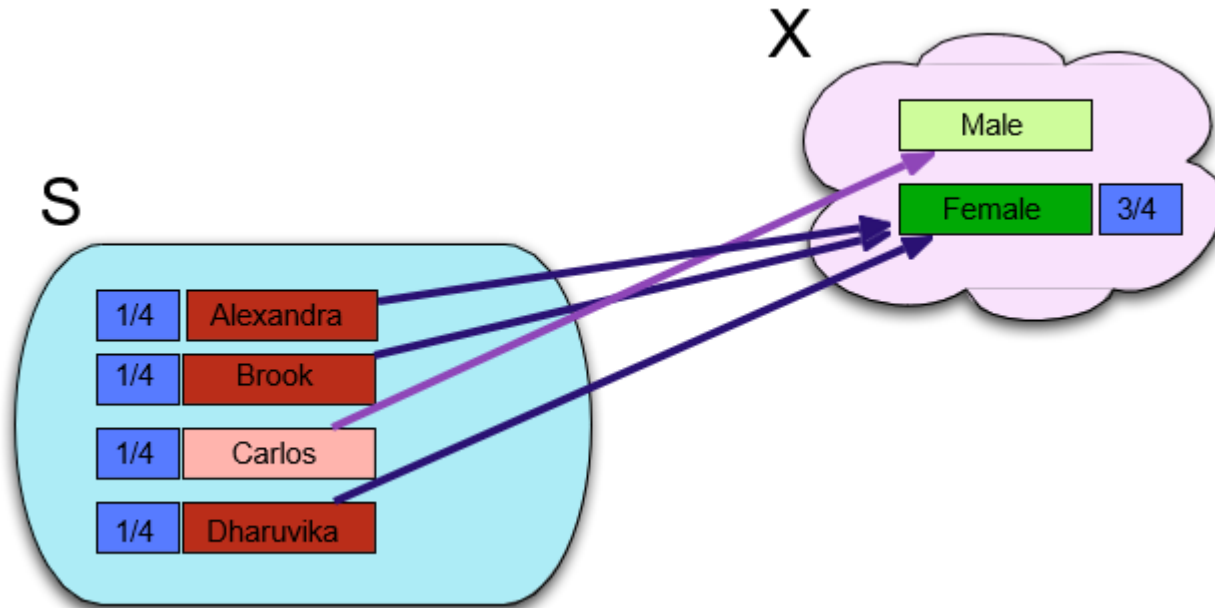
Discrete distributions

- How to find $P(X = \text{Female})$?



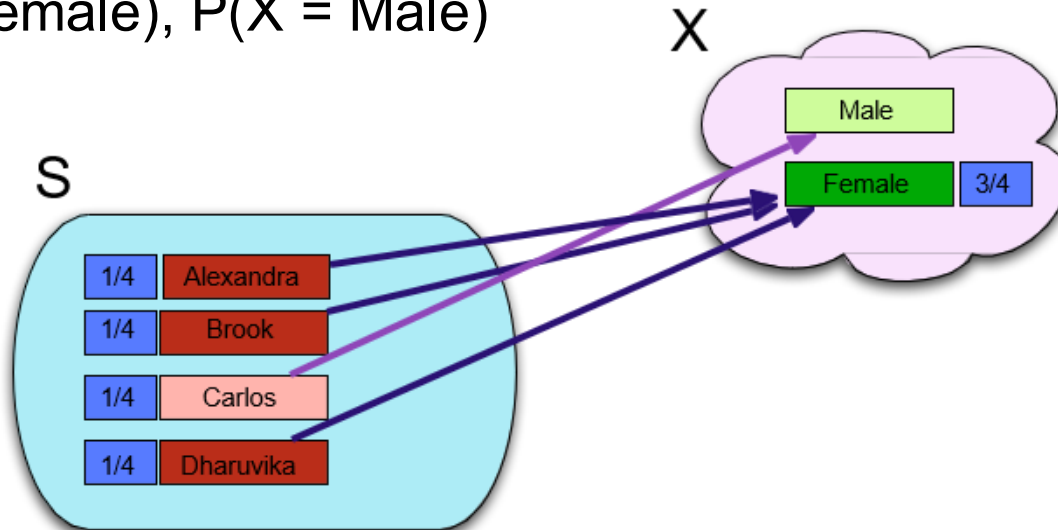
Discrete distributions

- How to find $P(X = \text{Female})$?



Discrete distributions

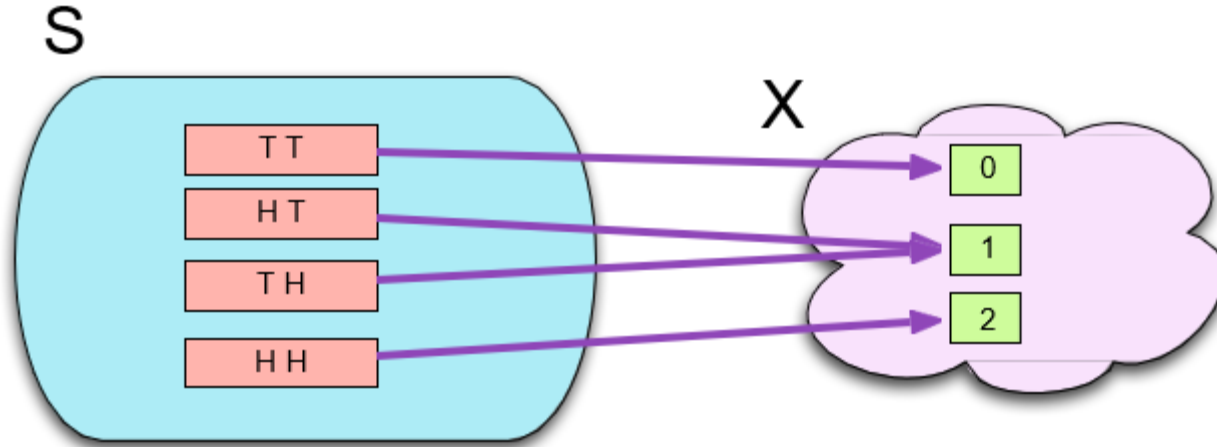
- What is the distribution of random variable X ?
 - $P(X = \text{Female})$, $P(X = \text{Male})$



x	Male	Female
$P(X = x)$	$1/4$	$3/4$

Discrete distributions

- What is the distribution of random variable X ?



x	0	1	2
$P(X = x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

Properties of Discrete Distributions

- We can write $P(X = x)$ to mean “The probability that the random variable X takes the value x ”.
- What must be true of these probabilities?

Properties of Discrete Distributions

1. Each $P(X = x)$ is a probability, so must be between 0 and 1.
2. The $P(X = x)$ must sum to 1 over all possible x values.

Probability Mass function (PMF)

The Probability Mass Function

A discrete random variable, X , can be characterized by its **probability mass function**, f (might sometimes write f_X if it's not clear from context which random variable we're talking about).

The PMF takes in values of the variable, and returns probabilities:

$f(x)$ is *defined* to be $P(X = x)$

PMF is a table

- Think of the PMF as a lookup table.

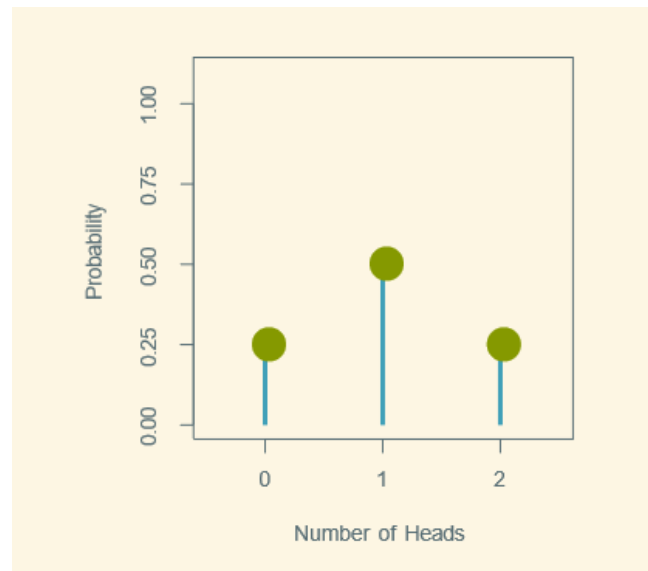
x	Male	Female
$P(X = x)$	$1/4$	$3/4$

- Best way to think of discrete random variables: they take various values, and each value has a certain probability of happening.

Visualizing discrete distributions: spike plot

Flip two coins at the same time, probability distribution of number of heads:

- Often use the spike plot
- Like a bar plot, but with probabilities, instead of frequencies or proportions, on the y-axis.



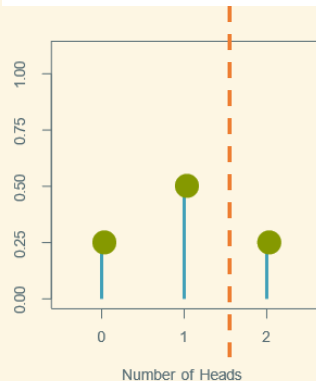
The cumulative distribution function (CDF)

- Often, we are interested in the probability of falling in some range of values.
- We can use the cumulative distribution function (CDF), which gives the “accumulated probability” up to a particular value.

The Cumulative Distribution Function

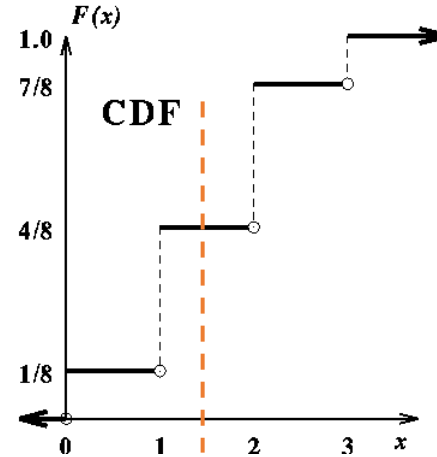
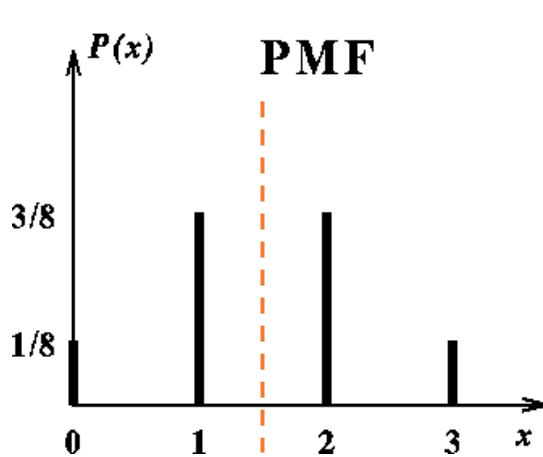
A random variable, X , can be characterized by its **cumulative distribution function**, F (or sometimes F_X if we need to be explicit), which takes values and returns *cumulative* probabilities:

$F(x)$ is defined to be $P(X \leq x)$



Relating PMF to CDF

- How can we calculate $F(x)$ from the PMF table f ?
 - Add up all the probabilities up to and including $f(x)$.
 - What is the value of $F(-0.1)$ (i.e., $P(X \leq -0.1)$)? $F(1.5)$?



- For discrete random variables, $F(x)$ *jumps* at locations with nonzero probability mass

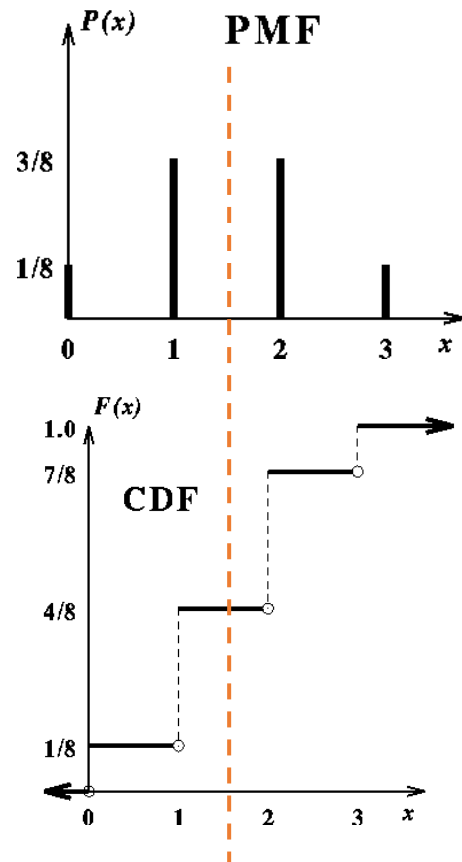
Relating PMF to CDF

- So the PMF of X is:

$$f(x) = \begin{cases} 1/8, & x = 0 \\ 3/8, & x = 1 \\ 3/8, & x = 2 \\ 1/8, & x = 3 \end{cases}$$

- We can write the CDF of X :

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{8}, & 0 \leq x < 1 \\ \frac{4}{8}, & 1 \leq x < 2 \\ \frac{7}{8}, & 2 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$



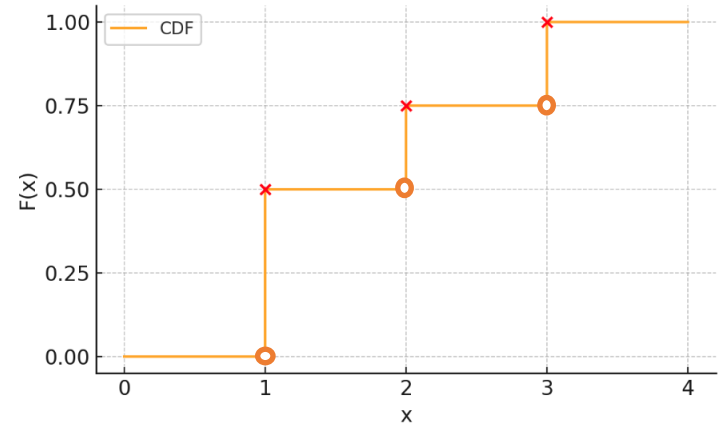
In-class activity

- Given by the PMF of X , find the CDF of X .

$$f(x) = \begin{cases} 1/2, & x = 1 \\ 1/4, & x = 2 \\ 1/4, & x = 3 \end{cases}$$

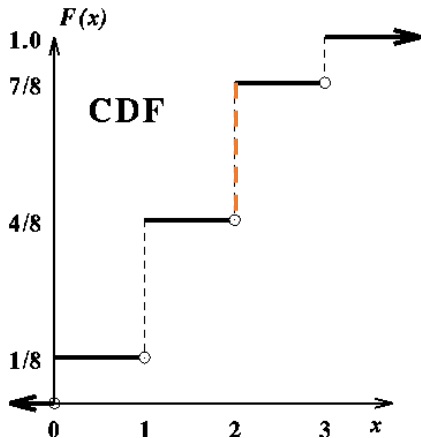
- Answer:

$$F(x) = \begin{cases} 0, & x < 1 \\ \frac{1}{2}, & 1 \leq x < 2 \\ \frac{3}{4}, & 2 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$



Relating CDF to PMF

- How could we find $f(x)$ from a cumulative distribution function F ? e.g., $f(2)$?

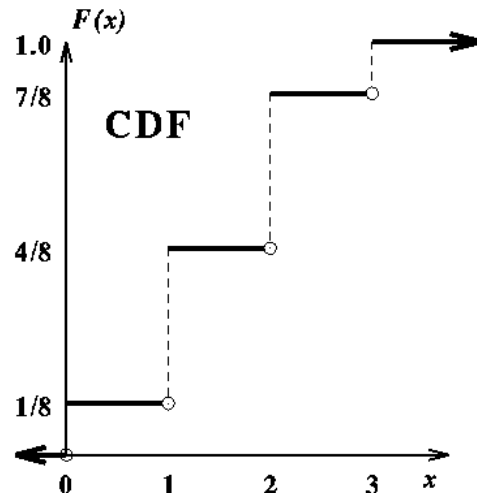


- Focus on “jumps”: $f(x) = F(x) - F(\text{jump just below } x)$
 - $f(2) = F(2) - F(1) = \frac{7}{8} - \frac{4}{8} = \frac{3}{8}$
 - $f(2.1) = F(2.1) - F(2) = \frac{7}{8} - \frac{7}{8} = 0$
 - $f(1.5) = F(1.5) - F(1) = \frac{4}{8} - \frac{4}{8} = 0$

Exercise: using CDF and PMF

Given the CDF F :

- How to calculate $P(X > x)$?
 - $P(X > x) = 1 - P(X \leq x) = 1 - F(x)$
- How about $P(X \geq x)$?
 - $P(X \geq x) = 1 - P(X < x) = 1 - (P(X \leq x) - P(X=x))$
 - $1 - F(x) + f(x)$
 - $f(x)$ can be 0 or nonzero, depending on whether x is a jump



Exercise: using CDF and PMF

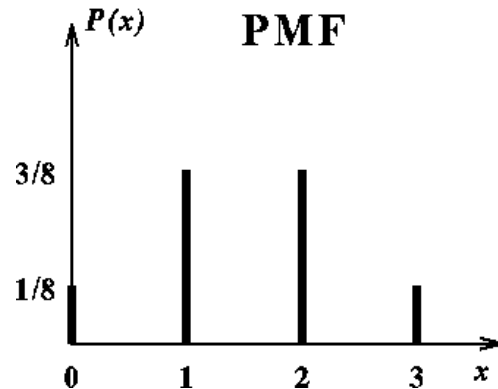
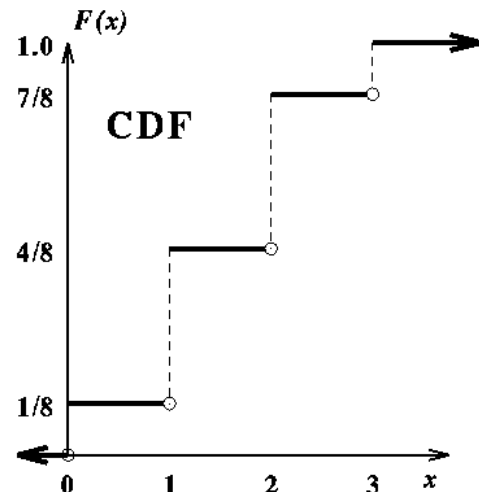
- What is $P(X \geq 2)$?
 - $P(X \geq x) = 1 - F(x) + f(x)$
 - $f(x)$ can be 0 or nonzero, depending on whether x is a jump

Using the formula:

$$\bullet \quad P(X \geq 2) = 1 - F(2) + f(2) = 1 - \frac{7}{8} + \frac{3}{8} = \frac{1}{2}$$

Another way:

$$\bullet \quad P(X \geq 2) = P(X = 2) + P(X = 3) = \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$$



Exercise: using CDF and PMF

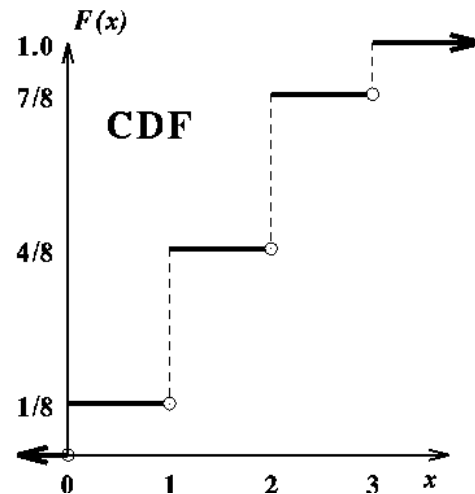
Given the CDF F :

- How to calculate $P(a < X \leq b)$?

$$= P(X \leq b) - P(X \leq a)$$

$$= F(b) - F(a)$$

- How to calculate $P(a < X < b)$?
 - (I'll leave this to you as an exercise..)



Transformations of random variables

- If X is a random variable, then $X + 5, 3X, X^2, \dots$, are all random variables
- Given any transformation function f , $f(X)$ is a random variable
- How to find the PMF of $f(X)$ based on that of X ?
 - First, find all values $f(X)$ can take
 - For each value c , try to find $P(f(X) = c)$

Examples

- Suppose X has PMF

x	1	-1
$P(X = x)$	0.5	0.5

- What is the PMF of $Y = X + 5$?
 - Y can take values 6 and 4
 - $P(Y = 6) = P(X = 1) = 0.5$
 - $P(Y = 4) = P(X = -1) = 0.5$

y	6	4
$P(Y = y)$	0.5	0.5